

Amir Mirzai Golpayegani

📍 Calgary, AB, Canada

✉ amirgolpaa24@gmail.com | ✉ amir.mirzaigolpayega@ucalgary.ca | ☎ +1 (587) 439-0436

🌐 [linkedin.com/in/amir-golpayegani-218148165](https://www.linkedin.com/in/amir-golpayegani-218148165) | 🐙 github.com/amirgolpaa24 | 🌐 Portfolio

Summary

M.Sc. Computer Science graduate with experience building Python/C++ systems for machine learning, NLP-adjacent backend services, model evaluation, data pipelines, and large-scale data processing. Skilled in PyTorch, Scikit-learn, REST APIs, SQL, Docker, Git, Linux, and reproducible experimentation, with applied research experience in retrieval, data infrastructure, LLM workflows, and AI automation.

Education

M.Sc. in Computer Science <i>University of Calgary</i>	GPA: 3.94/4.0	Sep 2022 – Feb 2026 <i>Calgary, Canada</i>
B.Sc. in Computer Engineering <i>Sharif University of Technology</i>	GPA: 3.69/4.0	Sep 2016 – Feb 2021 <i>Tehran, Iran</i>

Technical Skills

Programming: Python, C++, JavaScript, SQL

ML / AI: PyTorch, Scikit-learn, TensorFlow, CNNs, model training, evaluation, LLM workflows, AI agents

Data / NLP: NumPy, Pandas, SciPy, text processing, retrieval workflows, Google Earth Engine API, GDAL, Rasterio

Backend / Engineering: Django, Node.js, REST APIs, PostgreSQL, Docker, Git, Linux, AWS, GitHub Actions

Visualization / Automation: Streamlit, Polyscope, OpenGL, GLSL, n8n, OpenClaw, SearXNG, Telegram API

Work Experience

Research Software Developer <i>BigGeo / Mitacs, University of Calgary</i>	Sep 2022 – Feb 2026 <i>Calgary, Canada</i>
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- Built C++/Python software for scalable processing, storage, and retrieval of large Earth observation datasets.
- Automated SAR and optical imagery preprocessing using GDAL, Rasterio, SNAP, and satellite data APIs.
- Developed DGGs-based encoding pipelines to convert satellite imagery into structured tensor files.
- Improved encoding performance with C++ multithreading, reducing 10-tensor processing from 233s to 26s.
- Reduced storage by 38% for scaled Canada-wide data with 1,112 GeoTIFF tiles and 16K+ DGGs tensors.

Back-End Developer <i>Asr Gooyesh Pardaz</i>	Jun 2020 – Jul 2022 <i>Tehran, Iran</i>
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- Developed Django REST APIs for NLP, text-to-speech, and speech-to-text service plans backed by PostgreSQL.
- Built authentication features including registration, Google login, password reset, and email/phone verification.
- Implemented backend logic for payment-service integration and storing plan/payment records in a relational database.
- Contributed to a WebRTC-based chatbot system using GPT-style responses and FAQ data to answer client questions.

Deep Learning Teaching Assistant <i>Neuromatch Academy</i>	Jul 2026 <i>Remote, United States</i>
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- Mentored an international pod of ~15 students through Neuromatch Academy's intensive three-week Deep Learning course.
- Facilitated hands-on PyTorch tutorials spanning MLPs, CNNs, attention/transformers, diffusion generative models, and reinforcement learning.
- Guided project groups through end-to-end deep learning research projects, from question formulation and literature review to model implementation and final presentations.

Teaching Assistant <i>Department of Computer Science, University of Calgary</i>	Dec 2022 – Apr 2025 <i>Calgary, Canada</i>
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- **Developing Big Data Applications:** Mentored +25 students on AWS-based data pipelines, cloud storage, debugging, and scalable data workflows.
- **Working with Data at Scale:** Supported +20 students with large-scale data processing, data pipelines, and high-volume data analysis.
- **Database Management Systems:** Assisted +75 students with SQL, database design, course projects, grading, and exam support.

Research and Technical Projects

Agentic Job Search Automation Platform <i>OpenClaw, n8n, LLMs, AI Agents, Docker, Workflow Automation, SearXNG, Telegram API</i>	May 2026 – Present
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- Building an agentic AI system to search jobs, analyze job fit, and generate tailored application materials.
- Designed LLM-based workflows for resume/job matching, skill-gap analysis, and application decision support.

- Integrated Google Sheets, Google Drive, and Telegram API to track jobs and send automated application alerts.
 - Prototyped the initial workflow in n8n before migrating toward a more flexible OpenClaw-based agent system.
- DEM Super-Resolution Framework

May 2024 – Feb 2026

Python, PyTorch, CNNs, Scientific Imaging, Google Earth Engine API, Polyscope, Data Processing and Visualization

 - Trained deep ResNet models for 4× DEM super-resolution, reconstructing 30m elevation data from 120m inputs.
 - Built a Streamlit pipeline using Rasterio to parallelize GEE DEM tile downloads and collect mountainous terrain data worldwide.
 - Implemented PyTorch training and evaluation workflows with terrain-aware metrics such as RMSE, slope, TRI, TPI, and wavelet-based errors.
 - Analyzed model generalization across geographic regions and used evaluation results to compare reconstruction quality.

Spatiotemporal Earth Observation Data System Using DGGS

Sep 2022 – Feb 2026

First-author publication | M.Sc. thesis | C++, Python, DGGS, Wavelets, B-splines, Time-Series

 - Designed a DGGS-based framework for real-time multiresolution management of satellite imagery.
 - Developed a tensor-based storage model for direct access to arbitrary regions at multiple resolutions.
 - Built B-spline temporal approximation to compress time-series data and retrieve missing timestamps locally.
 - Retrieved 105,489 DGGS cells per time-series query in 13.9s, compared with about 220s using global retrieval.

Publication

Amir Mirzai Golpayegani, Mahmudul Hasan, Faramarz F. Samavati. *Real-Time Multiresolution Management of Spatiotemporal Earth Observation Data Using DGGS*. Remote Sensing, 2025. doi:10.3390/rs17040570

Certificates

Supervised Machine Learning: Regression and Classification, DeepLearning.AI and Stanford University, Coursera Oct 2024
Applied Machine Learning in Python, University of Michigan, Coursera Jul 2020

Languages

English: Full Professional Proficiency (TOEFL 115/120)
Persian: Native Proficiency

Achievement

Teaching Assistant Excellence Laureate 2024/2025
Department of Computer Science, University of Calgary — awarded for outstanding teaching support (link)
Ranked 17th Nationwide in Iran University Entrance Exam (Konkour) May 2016
Top 0.001% among all participants